Advanced Textile Sorting and Pre-processing Facilities: The Business Case

Research date: October – December 2024
Publication date: March 2025 (last updated May 2025)
Project code: VTU019-001 (ACT UK WP6.09)

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Document reference
(please use this reference when citing WRAP's work)
WRAP. (2025) 'Advanced Textile Sorting and Pre-processing Facilities: The Business Case'

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Acknowledgements

WRAP would like to thank Circle Economy for their expertise, thorough research and analysis that has informed this report, as well as Marieke Akerboom for developing the dedicated cost model:

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Circle Economy are a global impact organisation with an international team of passionate experts based in Amsterdam. They empower businesses, cities and nations with practical and scalable solutions to put the circular economy into action. Their vision is an economic system that ensures the planet and all people can thrive. To avoid climate breakdown, their goal is to double global circularity by 2032.

WRAP would like to thank the following Partners for their valuable feedback and insights towards this report:

circle 8 textile ecosystems	Cyndi Rhoades Judith Rosser-Davies Marc Lopresto
The Salvation Army Trading Company	Bernie Thomas Majonne Frost Nicole Morarescu
Shred Station	Kristian Carter
TRA Textile Recycling Association	Alan Wheeler Mohammed Patel
Textile Recycling International – Return, Renew, Reuse, Recycle	Peter Page
ukft	Adam Mansell Flor Canencia Jaki Love Lauren Junestrand

Finally, WRAP would like to acknowledge the valuable contributions of those who attended workshops, agreed to interviews, provided insights and support to all parts of Work Package 6 considered in this report:

Alison Carey	Chris Carey's Collections Ltd	Lieve Vanrusselt	Matoha
Emma Joy	Oxfam	Lisa Ly	Dunelm
Hannah Carter	ReSkinned	Louisa Hoyes	TOMRA
Jacks Guinness	WRAP	Marc Ratzker	Coppermill Ltd
Jessica Salt	LW Sait and Sons LLP	Micki Filocco	Tesco
John Clay	Edward Clay & Son Ltd	Moritz Grün	Texaid
John Roberts	Roberts Recycling	Parikshit Goswami	University of Huddersfield
Justin Guest	Archipelago Eco Investors	Rachel Singer	ReLondon
Katharine Beacham	Marks and Spencer	Robert Pretorius	Anglo Recycling Company Ltd
Katharine Fox	WRAP	Ross Barry	LMB Textiles
Krishiv Vyas	Fortune Eximports Ltd	Roy Harris	Rontex Ltd
Laila Ali	WRAP	Shan Dulanty	The Manufacturing Technology Centre Ltd
Laura Pearl	WRAP	Steven Bethell	Bank and Vogue

1.0 Introduction

Textiles recycling is widely accepted as the primary route to close the loop on materials. This is achieved by effectively managing non-reuseable textiles (NRT) volumes, maximising the use of resources, and supporting brands and retailers in meeting environmental targets by replacing virgin materials with recycled content. However, despite large volumes of NRT sent directly to end-of-life (EOL) streams by citizens or collected, NRT resources remain underutilised, due to the barriers in sorting for recycling. The current post-consumer textile (PCT) value chain relies almost entirely on manual labour and is not yet equipped to cater to emerging recycling solutions that require consistent, reliable feedstock that meet specific requirements. Advanced Textile Sorting and Preprocessing (ATSP) facilities show great promise in streamlining the sorting process through automation, reducing sorting costs and increasing input/output capacity and quality. Fundamentally aiming to bridge the gap between NRT supply and recyclers.

This report consolidates research on the feasibility and viability of implementing a standalone ATSP, envisioned as one of several facilities to meet recycled content demand and take responsibility of our NRT in the UK. For a macro view of ATSP networks within the broader textiles ecosystem, please read 'Transitioning to a UK circular textiles ecosystem'[1]. This report begins by presenting the challenge and opportunity of the current NRT system, examining the business-as-usual cost model, volumes of NRT and material flows. A review of the ATSP landscape follows, including potential ATSP configurations and technologies, as well as assessing the UK and EU textiles recycling market and optimal facility locations. The ATSP cost model is then introduced to test the economic viability of an ATSP, outlining key assumptions and costs, such as capital expenditure (CAPEX) and labour requirements. A summary of Circle-8 Textiles Ecosystems business case follows, demonstrating the opportunities that ATSPs can leverage to meet industry needs. The report highlights the potential of ATSP circular textiles infrastructure and concludes with opportunities to refine and adapt the cost model to create a profitable operating business proposition. It emphasises the crucial role of key enablers such as government, green investors and brand and retailers in supporting ATSPs and implementing the necessary market dynamics to establish effective recycling markets.

The ACT UK project

The Automatic-sorting for Circularity in Textiles (ACT UK) project aims to build a blueprint and business case for ATSPs in the UK. ACT UK brings together a diverse consortium of recyclers, collectors and sorters, academia, manufacturers, associations, brand and retailers and eco-system developers, to catalyse the formation of an evolved market supplying high volume, fibre and colour-sorted, pre-processed feedstock to existing and emerging mechanical and chemical recycling processes. The scope of the project has focused on finding solutions in the UK and Europe, with consideration to vital markets in wider geographies as automated sorting and textile recycling scales.

2.0 Current NRT profile

This section presents a market analysis of NRT in the UK, estimating the NRT fraction from total textile arisings and runs through the cost model of the current system. The NRT profile showcases the huge volumes of NRT being uncaptured and/or underutilised at present, clearly highlighting the potential of the material as an ATSP input. View 8.1 Appendix 1 for NRT profile methodology and sensitivity analysis.

2.1 NRT flows in the UK

See Figure 1 for diagram on annual NRT volumes arising in the UK and onward flows, as well as key stakeholders collecting material. Main findings[2]:

- **NRT total:** Estimated 744,000 tonnes (T) of NRT are discarded annually.
- **Residual waste:** The majority (52%/387,000 T) of total NRT is disposed of via residual waste streams by citizens, where it is destined for incineration or landfill. Capturing this would increase volumes of potential ATSP input material. This however would require substantial changes in citizen disposal behaviour and an optimised PCT collections system.
- **Collected:** 206,000 T of NRT is collected by the existing value chain.
- **Exported:** A significant portion of the NRT collected is exported as a resource for manual sorting (77% /158,000 T). Of these exports, the majority (92,000 T) is sent to Pakistan for remanufacturing or mechanical recycling. 44,000 T is sent to the United Arab Emirates (UAE), where the textiles will be further sorted.
- **Recycled:** Only a small fraction of NRT collected is currently recycled domestically and abroad. 27,000 T open-loop recycling in the UK and Europe and 1,000 T closed-loop.

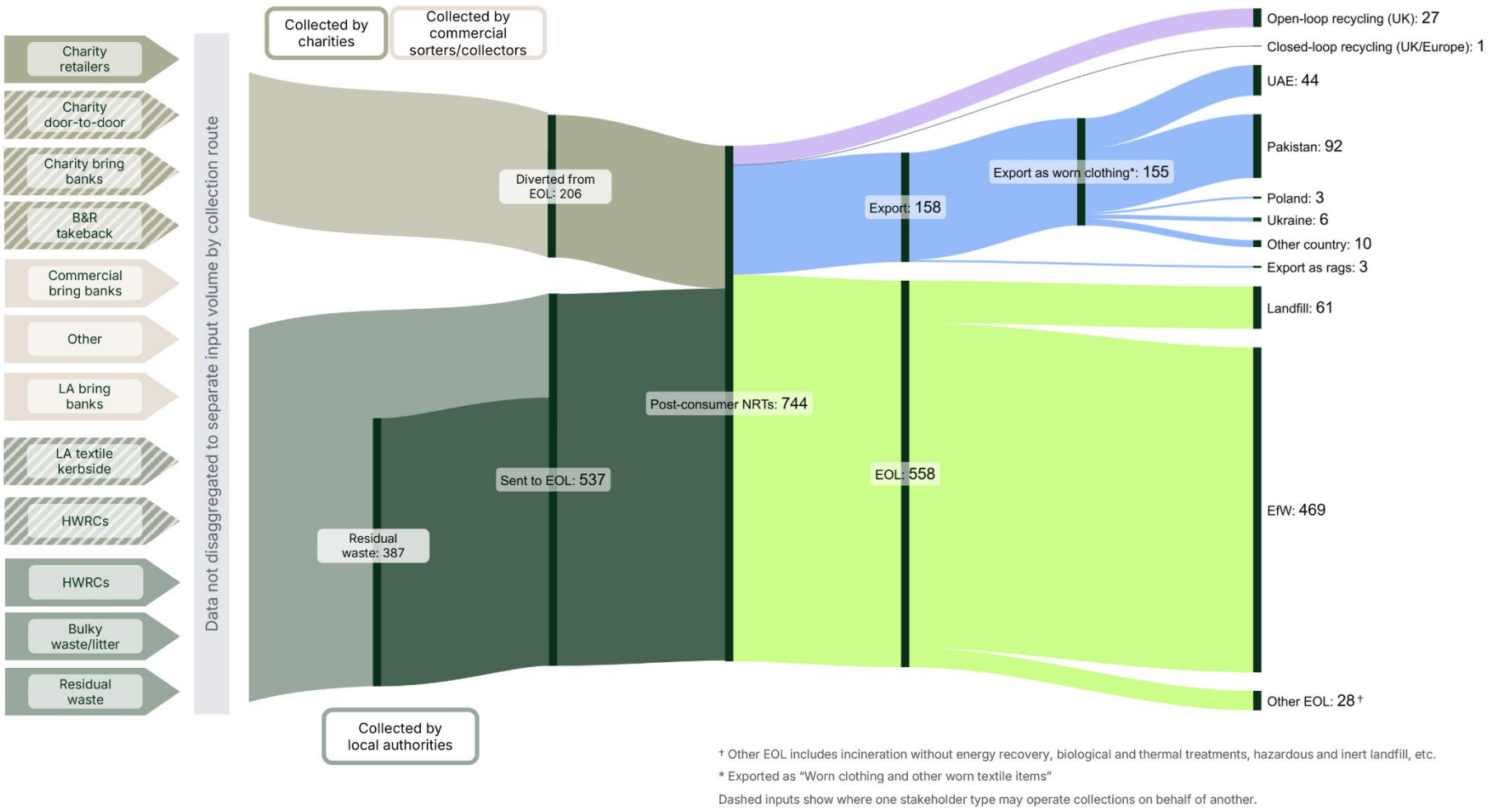


Figure 1: Current NRT flows annually in UK. All numbers expressed in 1000s of tonnes. Quantities are not disaggregated for most collections to avoid double counting.[2]

2.2 Potential NRT ATSP input material

Material input to ATSPs could reach 335,000 T by 2035[3]. This figure is based off the most likely scenario in which, compared to 2024 rates, volumes of PCT generated remain stable, collection rates moderately increase, and citizens dispose of the same levels of textiles into residual bins. The 335,000 T would require 14 ATSPs to handle NRT material input at 25,000 T per annum. For further reading on scenarios that investigate the potential landscape for NRT inputs in the future (2035), please read *Transitioning to a UK circular textiles ecosystem*.[1]

Although professional textiles were outside the scope of the ACT UK project, they could be utilised as immediate inputs to ATSPs as facilities scale up. ATSPs provide a possible path for professional textiles moving up the resource hierarchy, bypassing end-of-life routes that are common due to business sensitivities and/or health and safety standards. Additionally, professional textiles have uniform characteristics such as material composition and colour, allowing for more streamlined processing at ATSP facilities.

2.3 Current NRT cost model

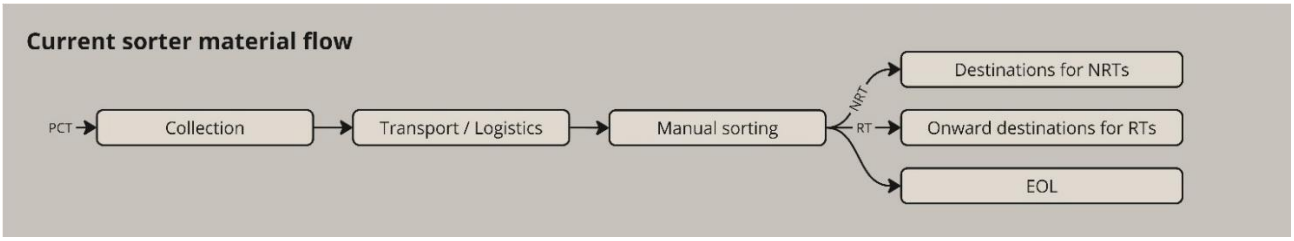


Figure 2: Current material flow of PCT.

Collections, transport/logistics and manual sorting cost a sorter on average £526 per tonne[4] under current processes (see Figure 2) for standard textile flows. If this material is then sold at an average NRT value of £100 per tonne, the NRT stream alone generates a loss of £426 per tonne (see Figure 3). When extrapolating this to the 206,000 of NRT[2] collected in the UK every year, this translates to around £88 million[1] related to NRT sorting costs.

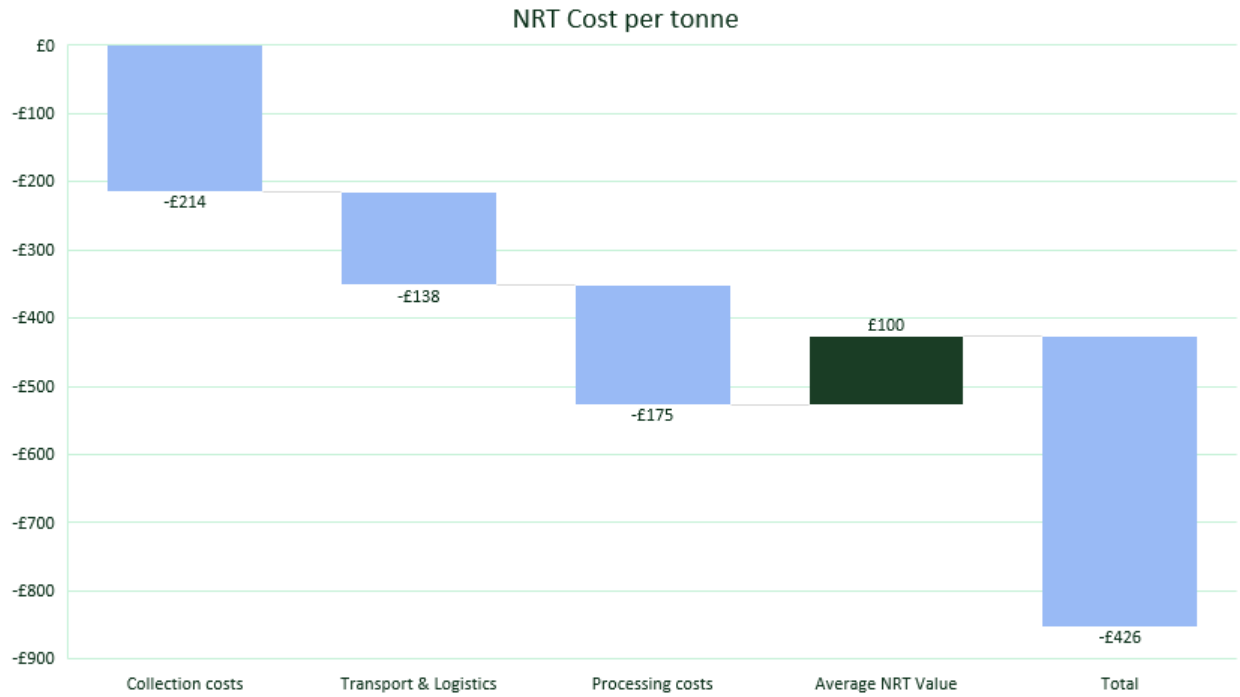


Figure 3: Current NRT cost overview under manual sorting.

Whilst RT sales have historically secured revenue and compensated NRT losses, industry experts have flagged shrinking value and volumes of RT, is exposing sorters to increased financial burdens. Pressures on RT further highlight the need for new business models, particularly around NRT processing and ATSPs.

3.0 ATSP landscape review

ATSPs are a missing link in the current textiles value chain - see Figure 4, where ATSPs have been mapped on to an ambitious vision for a circular textiles ecosystem. By providing an onward NRT market for collectors and sorters, ATSPs have the potential to support the flow of NRT and transform volumes into recycler feedstock. To understand ATSPs role in embedding circularity, this section outlines ATSP configurations, technologies to utilise and potential locations, as well as summarising the current recycling market and recycler requirements.

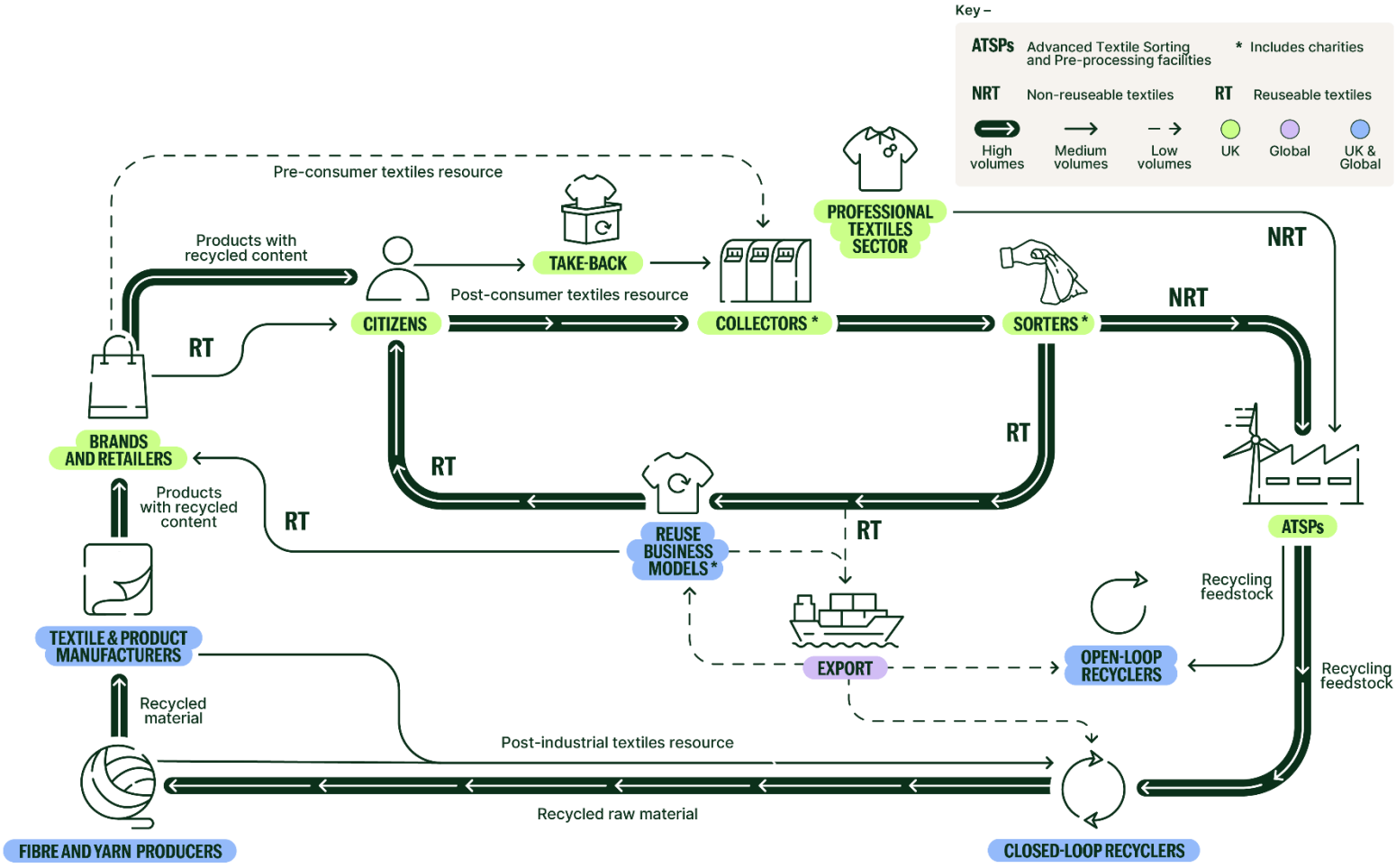


Figure 4: Diagram of an ATSP integrated circular textiles value chain vision including key stakeholders and movement of RT/NRT.

3.1 ATSP Value Chain configurations

It is important to highlight that ATSPs are not confined to a specific configuration or business model. Flexibility and creativity in ATSP configurations should be utilised to suit the business needs and future vision of textiles value chain stakeholders. See Figure 5 for a non-exhaustive list of ATSP configurations. Note, 'sorter' in the figure refers to any stakeholder conducting the sorting of textiles. This can include, but not limited to collectors, charities, waste management companies and brand and retailers.

The cost model in section 4.0 is based on a generic, standalone, 'Sorter to ATSP' ownership model and therefore follows specific assumptions relating to transport/logistics, operating costs and CAPEX of a single facility. In the ownership model, textiles are purchased by the ATSP at a cost and become its direct responsibility. The ownership model is sensitive to market fluctuations such as recyclable feedstock demand, input and output prices, as well as input quality provided by its suppliers.

A service model – outlined in Figure 5, is a far less market-sensitive model. In this case, the ATSP does not own the textiles processed, but rather provides a service by sorting, processing and sending the material to a recycler. Opportunity in the service model is highlighted in section 6.0.

Shared ownership could also optimise the operations of an ATSP, by vertically integrating it into existing solution providers, such as sorters and/or recyclers. This would offer several synergy opportunities on staffing, rent and utilities, as well as transport.

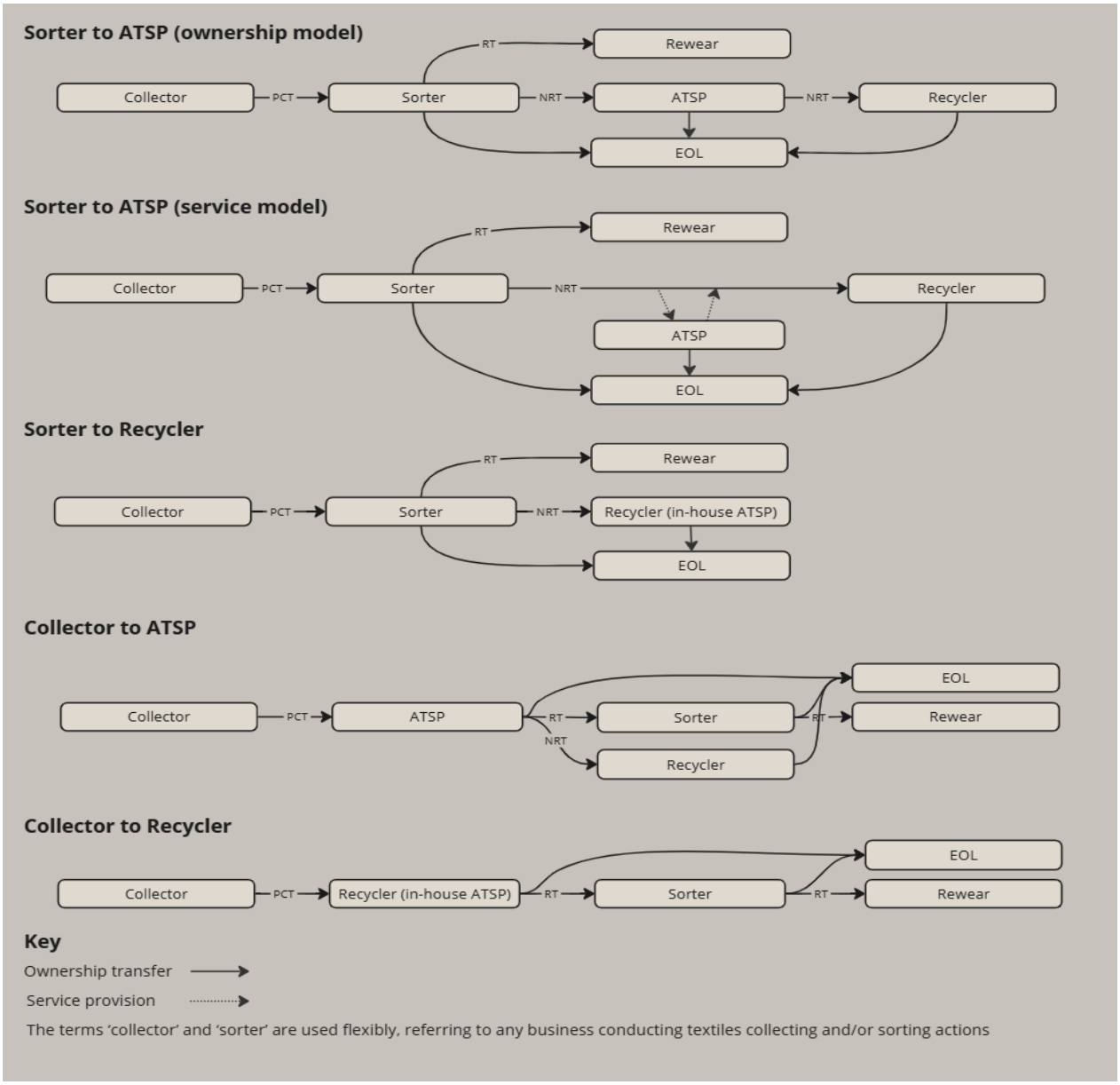


Figure 5: Possible value chain configurations.

3.2 ATSP technology landscape

Six automated technology providers are compared in Table 1: Automated sorting technology due diligence (information as of September 2024). All providers offer Near-infrared (NIR) Spectroscopy sorting solutions, sometimes coupled with image recognition solutions. All businesses can identify the most commonly recyclable fibres: cotton, polyester and wool, and could expand fibre type given research and development (R&D) funding. Current technologies vary in their capacity. A couple of large commercial solutions are reported to handle between 2.5 to 12.5 tonnes/hour. A handheld technology listed in the Table can process 100 kg/day per person, this technology is able to sort greater volumes than manual sorting (80 kg/day per person).[5] Note, this report has not researched direct comparisons between manual and automated sorting capacity. Regardless of manual versus automation assessments, NIR scanned feedstock is increasingly a requirement from textiles recyclers, therefore some level of automation will be necessary for sorting for recycling.

Table 1: Automated sorting technology due diligence (information as of September 2024).

Technology provider	Material composition sorting	Shape sorting	Colour sorting	Separation of disruptors	Material cutting	Technology specifications	Product offerings
ANDRITZ (ANDRITZ GROUP)	Yes	Yes	Yes	Yes	Yes	NIR	Emerging technology
Matoha (Matoha)	Yes	No	No	No	No	NIR	Handheld devices
PICVISA (PICVISA)	Yes	Yes	Yes	Yes	Yes	NIR & VIS	Handheld and large commercial solutions
REDWAVE (REDWAVE)	Yes	Yes	Yes	Yes	Yes	Not specified	Emerging technology
TOMRA (TOMRA)	Yes	Yes	Yes	Yes	Yes	NIR & VIS	Large commercial solutions
VALVAN (VALVAN)	Yes	Yes	Yes	Yes	Yes	NIR & RGB camera	Small and large commercial solutions

NIR technology limitations

NIR-based technologies enable efficient recognition of textile composition; however accuracy limitations exist. These limitations can reduce the acceptance rates of input materials at ATSPs, described in section 4.5:

- **Composition**
  - Blends – Often recognises up to two dominant fibre types.
  - Specific fabrics – NIR struggles with certain fabrics e.g., linen, leather. Limited in detecting elastane which is common in textile blends.
- **Surface scan** – Often only effective for monolayer garments as scanners and image-recognising technologies scan the top surface[6]. Thickness can also reduce accuracy.

- **Colours** – Detection is more effective with light and single-coloured textiles colours, but is more difficult with darker textiles and unable to recognise fibre composition dyed with carbon black ink. Ongoing R&D aims to improve this.[7]
- **Coatings/finishes** – Presence can reduce accuracy e.g., polyurethane coated materials
- **Moisture** - Liquids interfere with NIR spectrum analysis, and wet textiles must be removed to prevent contamination and enable accurate NIR processing. Some technologies offer blower points which can aid drying, whilst other remain reliable with a 0-2% moisture content[8].

Investment, innovation and R&D in NIR and other automated sorting technologies is actively taking place, paving the way for further advancements and optimisation in the sorting space.

3.3 ATSP customer requirements

Recyclers are currently the primary customers of ATSPs and effectively dictate specifications of ATSP outputs. Refer to Table 2 to view common recycler requirements of mechanical and chemical (open and closed-loop) feedstock. See Table 2 for detail on current specifications that can be used for ATSP output planning and revenue stream modelling. Read Manufacturing Technology Centre's (MTC) *'Condition of supply pre-processing technology assessment'*[9] for further information on recycler requirements.

Table 2: Common recycler requirements for chemical and mechanical recycling (open and closed-loop) feedstock.

Recycler requirements	Mechanical recycling (open and closed-loop)	Chemical recycling (open and closed-loop)
Material composition	Composition range dependent on recycling technology	Composition range dependent on recycling technology
Physical disruptors: Metal and plastic hardware	Removed for recycling	Removed for recycling
Physical disruptors: Other e.g., labels, embroidery	Recyclable	Recyclable
Physical disruptors: Coatings e.g., glitter, rubber, permanent press finishing	Non-recyclable	Non-recyclable
Chemical disruptors: e.g., optical brighteners, water-resistant finishing	Non-recyclable	Non-recyclable
Multi-colour	Open-loop recyclable Closed-loop non-recyclable	Colour range dependent on recycling technology
Multi-layers	Non-recyclable *Same composition recyclable	Non-recyclable *Same composition recyclable

3.4 ATSP customer landscape

Information on open and closed-loop recyclers in the UK and EU have been aggregated from WRAP's Textiles Sorting and Recycling Database[10], to highlight the growing geography and number of recyclers that ATSPs will supply to. See Figure 6: Map of all open/closed-loop recyclers broken down by recycling process listed in WRAP's Textiles Sorting and Recycling Database in the UK and EU. Find key in lower right corner. Table 2 or find full table in Table 11, Appendix 2. The number, technology and capacity of recyclers will grow as brands and retailers replace virgin material with recycled content to meet environmental targets, such as UK Textiles Pact[11] and Net Zero[12], and follow minimum recycled content policies.[13]

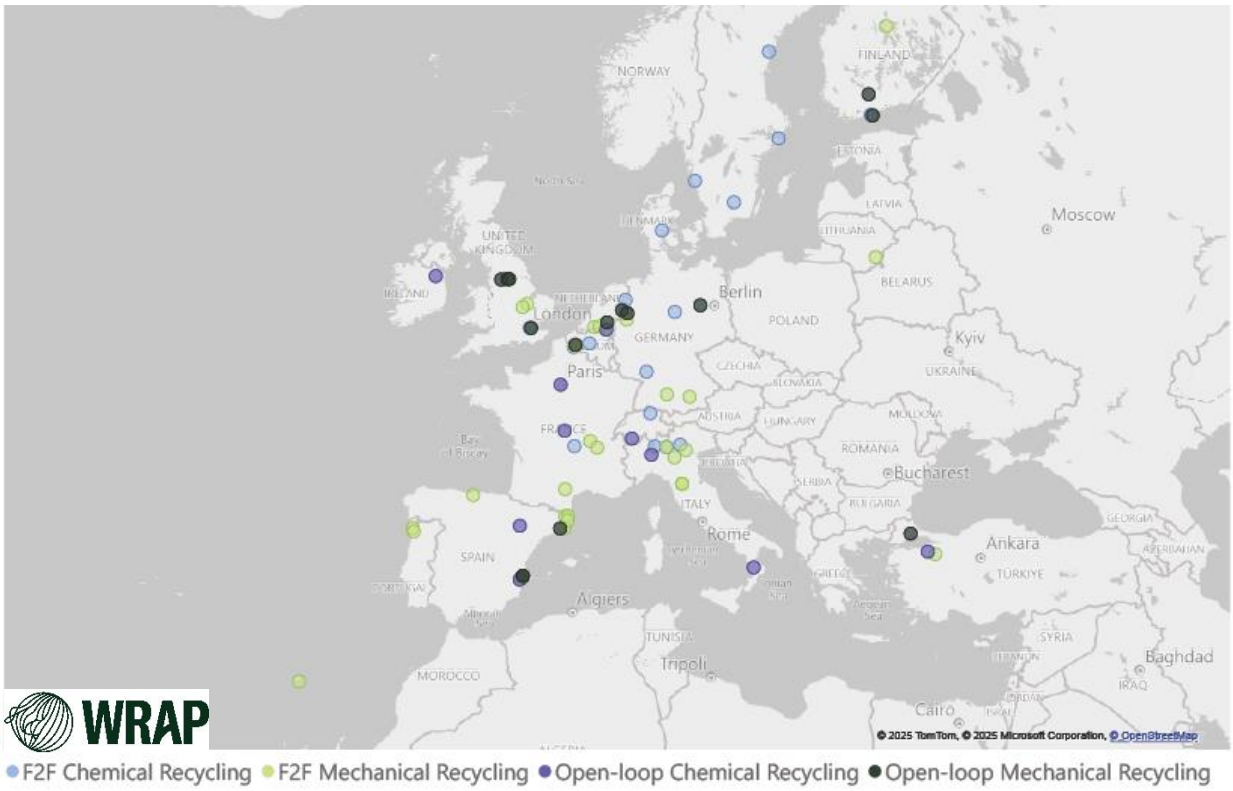


Figure 6: Map of all open/closed-loop recyclers broken down by recycling process listed in WRAP's Textiles Sorting and Recycling Database in the UK and EU. Find key in lower right corner.

In terms of scale, closed-loop mechanical recycling solutions are fully operational. Chemical recycling of cotton and polyester is subject to intense R&D and at the brink of commercialisation.[14] At present, cotton is the primary input material by volume for both recycling technologies, followed by wool and/or polyester. A number of recyclers also accept materials such as down, leather, acrylic, aramid, cashmere, polyamide. Capacity is an important parameter as it indicates potential market volumes for an ATSP, however data is not well defined. In the UK, businesses declaring their facility capacity total 9,000 tonnes of input per annum (T/a) at present and expect to process 1,000,000 T/a in 2030.[15] In the EU, declared capacity runs at 871,400 T/a, and is expected to reach 1,142,600 T/a in 2030, a 23% increase. The forecasted increase, from the point of research to 2030, further underscores the expected growth in the recycling sector and subsequently the need for feedstock.

Table 3: Information from WRAP's Textiles Sorting and Recycling Database[10] on open and closed-loop mechanical and chemical recyclers in the UK and EU .

	Open and closed-loop mechanical recyclers	Open and closed-loop chemical recyclers
Total recyclers	64	38
TRL level	27 At scale 3 Pilot/research	2 At scale 11 Pilot/research
Resource input	23 Post-industrial 18 Post-industrial/pre-consumer 1 Pre-consumer 10 Post-consumer 11 All inputs	10 Post-industrial 2 Post-industrial/pre-consumer 9 Post-industrial/post-consumer 8 Post-consumer 8 All inputs
Primary input material	Cotton	Cotton
Secondary input material	Wool/wool blends	Polyester

3.5 ATSP potential locations

Transport and logistics, as well as proximity to primary stakeholders, are important factors for ATSP success, helping to enhance operational efficiency, minimise costs and reduce environmental emissions.

Figure 7 displays potential ATSP locations within close distance to ports, densely populated areas and existing sorters. UK's 48 Special Economic Zones[16] are not mapped, however their locations largely correspond to hotspots identified. Cost of land has not informed research.

The most plausible ATPS locations identified are:

1. Midlands: proximity to Birmingham and Nottingham
2. North West: proximity to Liverpool and Manchester
3. South: proximity to London.

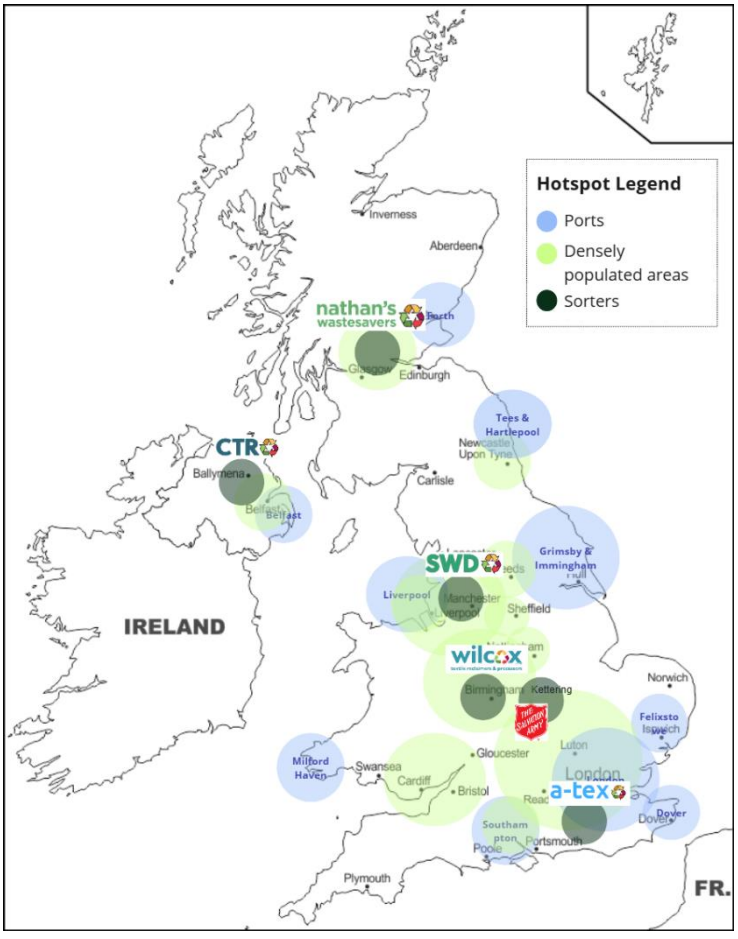


Figure 7: Hotspots for plausible locations of ATSPs around the UK.

4.0 ATSP operational assumptions & cost model

To fully understand the operational and financial sensitivities and opportunities of an ATSP, a model has been created for a generic, stand-alone, for-profit ATSP. Figure 8 visualises the key operational requirements and processes of an ATSP laid out in this chapter.

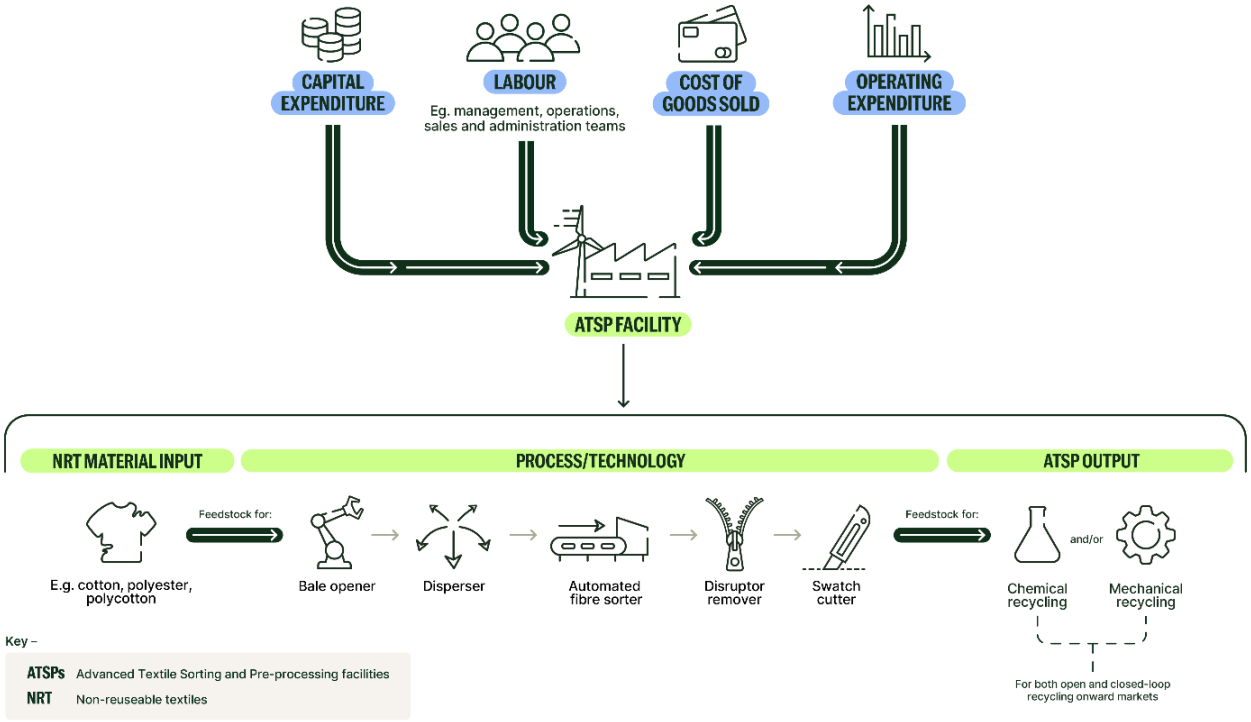


Figure 8: Key operational requirements and processes of a standalone, generic ATSP outlined in the cost model

4.1 ATSP operational assumptions

Operational assumptions are important in providing accuracy to the cost model in reflecting potential business activities. The assumptions can also inform decision making and identify areas of uncertainty and risk. Table 4 provides key cost and operational assumptions - see Table 12 in 8.3 Appendix 3 for details. It is important to highlight, the cost model is based off an ATSP with no external financial support including grants, investment and/or Extended Producer Responsibility (EPR) policy. Note, there is no current UK textiles EPR policy and its implementation remains uncertain.

Table 4: List of assumptions applied in the financial model of a generic, standalone, for profit ATSP.

Key input	Assumption
Corporate set up	Stand-alone facility
Financial support (e.g. grants, EPR)	None
Operational capabilities	Fibre sorting, disruptor removal and material size reduction
Operating capacity	Year 1 starting capacity: 10,000 T/a, Year 3 full capacity: 25,000 T/a
Facility cost	Rent, business rates, insurance, environmental monitoring, utilities, maintenance & repair
Start-up cost	Cost allocated to set up the facility in the first two years of its operations
Staff & staffing schedule	Year 1: 30 full time employees (FTEs), Year 3 (full capacity): 53 FTEs 3 shifts, 5 days a week, 260 days, 52 weeks per year
Transport cost structure	Transport from a supplier to the facility covered by the ATSP Transport from the ATSP to a customer not covered by the ATSP
Input material prices	Input prices are based on 4 different scenarios detailed below in Table 6
Output product prices	Product prices are split into 7 product categories detailed below in Table 5
Waste criteria	Cutting waste from the disruptor removal process and materials that do not fit sorting criteria are diverted to EOL

4.2 ATSP transition period

The transition from the current NRT flows – see 2.1, to a circular textiles ecosystem with an at-capacity network of ATSPs is expected to develop gradually. The ATSP modelled reaches full capacity 25,000 T/a after three years, starting at 10,000 T in year one. During this transition period, ATSPs will need to cater to intermediate markets.

In this phase, ATSPs should respond to market needs by marketing and selling to existing open-loop solutions, such as wipers and insulation. In the model, 75% of products are sold to open loop markets in year 1, reducing to 20% in year 3 – see Figure 9. Product prices for these NRT streams are lower than prices for closed-loop products - see Table 5, however, will serve to derisk existing open-loop markets and enable ATSPs to build stable off-take agreements with emerging closed-loop recyclers.

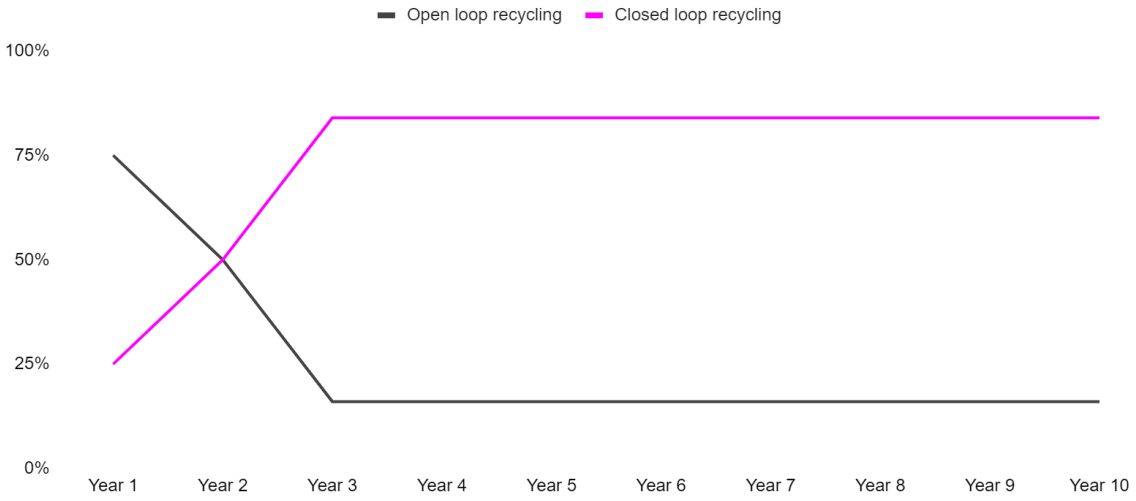


Figure 9: Indicative illustration of the transition from open to mainly closed-loop recycling product sales of an ATSP.

4.3 ATSP capital expenditure requirements

ATSP capital expenditure mainly includes technology investments – see 2.1. for key machinery. A 20% CAPEX contingency has also been applied, following industry practice. Every 5 years, a CAPEX upgrade is scheduled at 20% of the initial CAPEX investment.

The total capital expenditure costs of all the machinery mentioned varies depending on technology provider and ATSP configuration, this ranges between £15-22M.

4.4 ATSP staffing requirements

The ATSP requires skilled staff to be able to operate effectively. The modelled ATSP has 4 teams, with 30 staff members in year 1, increasing to 53 full time employees at full capacity. This number can vary greatly, particularly relative to an ATSP's business model configuration, capacity and technology. Salary ranges are applicable to the skill levels required for each role - See Table 13 in 8.3 Appendix 3 for further detail.

Management Team - Team of 3, responsible for guiding the company's strategic direction and ensuring operational efficiency. This includes the Managing Director, HR Manager, and Financial Controller.

Operations Team – Team of 8, ensures the facility runs smoothly and efficiently. This includes the Production Manager, Production Workers, Warehouse Manager, Warehouse Workers, Environmental and Industrial Risk Advisor, Engineer, Bale Press Worker, and Logistics Manager.

Sales Team - Team of 2, focuses on generating revenue and business growth. This includes the Sales Agents and the Business Development Manager.

Administration Team - Team of 1 to record and maintain financial transactions for the business, includes a Bookkeeper.

4.5 ATSP material and quality sensitivity

ATSP input material

Figure 10 shows the estimated NRT composition to be fed into an ATSP[17]. All other fibres and blends accounts for 28% of incoming material, this includes compositions that are not included in Figure 10 such as non-recyclable fibre blends. Material composed of >95% pure cotton represents 24% of incoming material, whilst textiles made up of >95% polyester constitutes 23% of the total material input. Materials with high compositions of cotton, wool and polyester are at or near market for F2F recycling – see section 3.4. These fibre types total 54% of overall ATSP input material, showing significant potential as F2F recycling feedstock (>95% cotton, >80% cotton, >95% polyester and >80% wool).

Please note that the estimated composition is indicative only, based on a single facility and relatively small sample size (5 T), and should not be considered representative of all NRT volumes. The material composition includes clothing and home textiles only, excluding accessories and shoes.

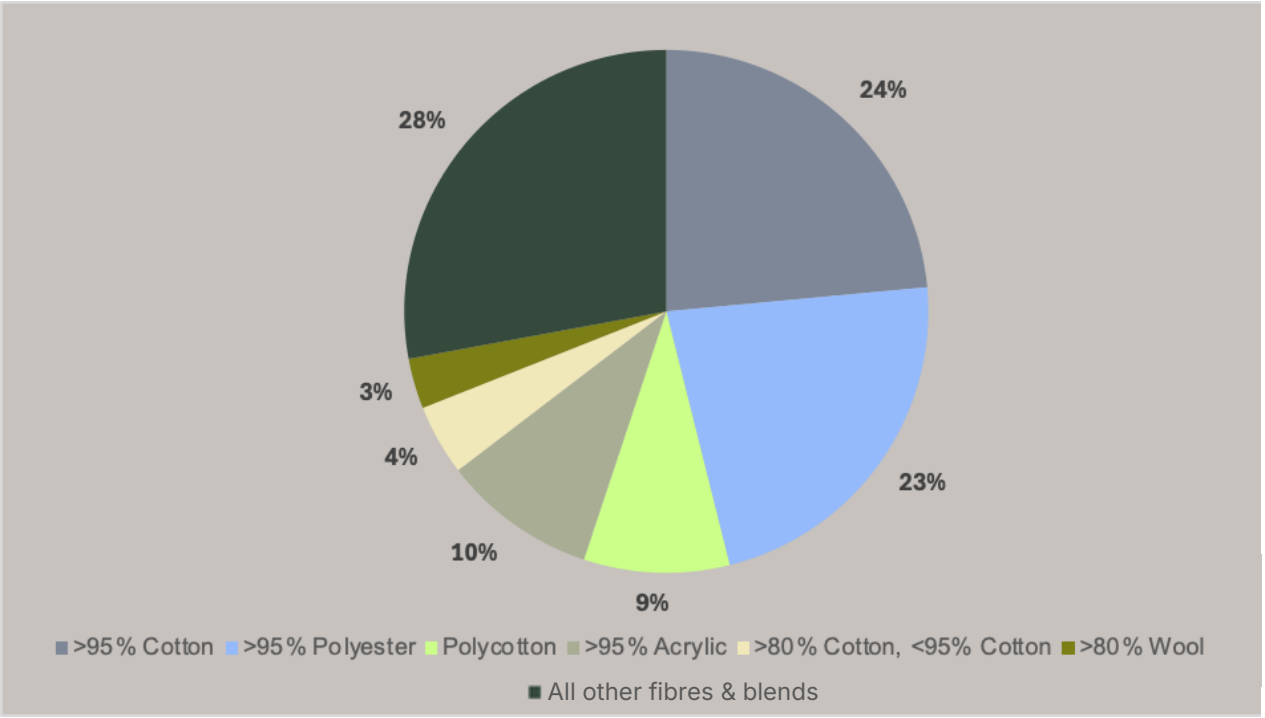


Figure 10: Indicative material composition of NRTs in the UK.

ATSP output material

The ATSP has seven different output grades - see Table 5. Three grades for chemical recycling, three grades for mechanical recycling and one grade tailoring to open-loop recyclers. Note, output grades[17] are based on current and near future recycler requirements and the material composition of NRTs in the UK. Table 5 also shows the estimated price at which ATSP outputs are expected to sell for[18]. The range varies depending on onward market

and recycling process, from £75 per tonne for open-loop markets to £1,200 per tonne for mechanical recycling of wool. Note, figures are indicative only. Gathering specific data on prices from stakeholders is challenging because closed-loop recycling is not yet widely at scale. Consequently, prices are not fully established and prices can vary greatly. Data may also be considered business sensitive.

Table 5: ATSP proposed output grade information.

ATSP output grade	Feedstock composition criteria	Disruptor presence criteria	Colour criteria	Output sales price in model
F2F Chemical Recycling - Polyester	≥95 polyester, <5% other fibres	No disruptor should be present	All	£200 per tonne
F2F Chemical Recycling - Polycotton	≥60% polyester, <40% cotton, <10% others	No disruptor should be present	All	£200 per tonne
F2F Chemical Recycling - Cotton	≥95% cotton, <5% other, no protein-based fibres	No disruptor should be present	All	£200 per tonne
F2F Mechanical Recycling – Cotton	≥80% cotton	Removable disruptors only	All monocolours[19]	£400 per tonne
F2F Mechanical Recycling - Acrylic	≥95% acrylic	Removable disruptors only	All monocolours	£90 per tonne
F2F Mechanical Recycling - Wool	≥80% wool	Removable disruptors only	All monocolours	£1,200 per tonne
Open-loop recycling	Criteria that satisfy current open loop markets			£75-150 per tonne

Acceptance rate

Figure 11 displays the ATSP material flows used in modelling.[20] The ATSP set up includes opening (Bale opener), evening out (Dispenser), and sorting materials automatically by fibre composition (Automated Fibre Sorter). NRT is sent to processing lines for 4 of the 7 output grades presented in Table 5 – note outputs grades are used as examples only to showcase where material is lost. A proportion of ATSP input material will not meet recycler feedstock criteria and will therefore be rejected, this percentage is known as the 'acceptance rate'. Figure 11 visualises and outlines the percentages at each stage of the process where textiles may be rejected from an ATSP. Multi-layered textiles are assumed to make up 15%[21] of total NRT, whilst NRTs with disruptors present could amount to 40%[21]. The financial performance of ATSPs are highly sensitive to the acceptance rate and volumes lost can significantly reduce potential revenue if onward markets are not found and the material is directed to EOL. There are opportunities to increase acceptance rates by setting offtake-agreements with suppliers to specify desired quality standards of NRT material and selecting high accuracy sorting equipment as technology advances.

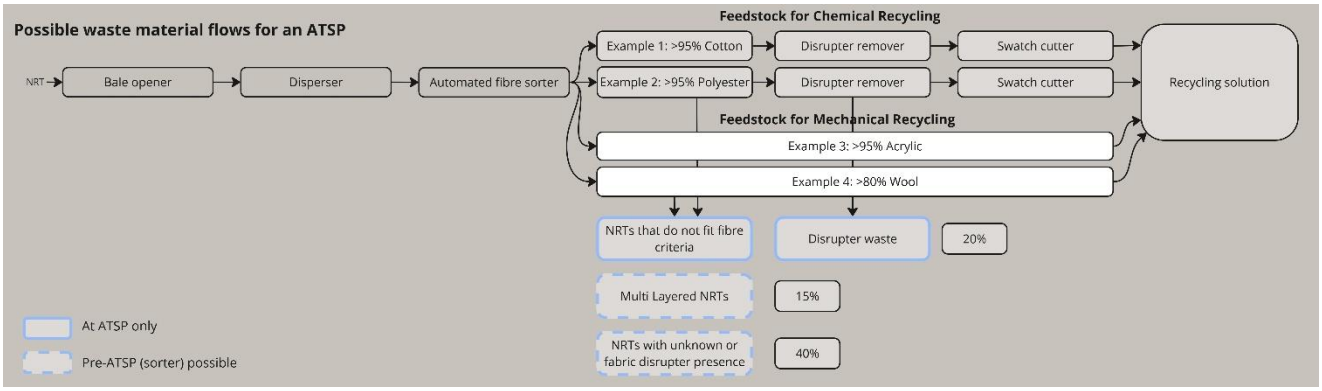


Figure 11: ATSP NRT material flows and rejected volumes.

4.6 Cost model scenarios

To test and scope the financial performance of an ATSP, the cost model employs four scenarios displayed in Table 6. In each case, ATSPs are responsible for transportation costs of input material only.

Table 6: NRT input costs per NRT input quality and its anticipated acceptance rates.

	Scenario 1: All NRT as revenue	Scenario 2: All NRT as cost	Scenario 3: Single layered NRT	Scenario 4: Single-layered NRT with removable/no disruptors present
Transport costs (per tonne)	£-138	£-138	£-138	£-138
NRT price (per tonne)	£-100	£-100	£-120	£-150
EfW gate fee	£-112	£-112	£-112	£-112
Multi layers present	Yes	Yes	No	No
Disruptors present	Yes	Yes	Yes	Only removable ones
Acceptance rate (% of total volume that is recyclable NRT)	46%	46%	54%	58%

4.7 Cost of goods sold (COGS) and operating expenditure

The cost of goods sold includes costs for start-up, input feedstock transport, feedstock purchase, labour and end-of-life treatment. It is assumed that all disruptor removal waste and all textiles that do not fit the sorting criteria will be sent to EOL treatment. Operating expenditure consists of overhead (non-COGS allocated), and facility costs such as ground rent, utilities, insurance, repairs and maintenance.

Earnings before interest, tax, depreciation and amortisation (EBITDA)

Table 7: Annual financial results of an ATSP at full capacity based on four scenarios.

	Scenario 1: All NRT as revenue	Scenario 2: All NRT as cost	Scenario 3: Single layered NRT	Scenario 4: Single-layered NRT with removable/no disruptors present
Revenue	£ 3,786,169	£ 3,786,169	£ 4,454,316	£ 4,820,075
Cost of goods sold	£-4,131,809	£-9,131,809	£-9,452,333	£-10,104,084
Gross Profit	£-345,640	£-5,345,640	£-4,998,017	£-5,284,009
Operating expenditure	£-2,250,211	£-2,250,211	£-2,250,211	£-2,250,211
EBITDA	£-2,595,852	£-7,595,852	£-7,248,228	£-7,534,221

Following the assumptions laid out, none of the scenarios described in Table 7 create positive EBITDA. Losses range from £7.6M to £2.6M annually at full capacity – see Table 8. This range is dependent on operating plan including staffing, transport costs, EOL costs, quality and availability of input feedstock, input material price, output feedstock price, demand for output feedstock and the acceptance rate. Whilst the negative figures highlight the evident need for financial support, the cost model also outlines several strategies to enhance the financial viability of the business case, which are discussed in section 6.0.

Table 8: Annual financial results of an ATSP at full capacity based on four different scenarios as described above.

	Scenario 1: All NRT as revenue	Scenario 2: All NRT as cost	Scenario 3: Single layered NRT	Scenario 4: Single-layered NRT with removable/no disruptors present
Revenue				
>95% cotton	£ 1,110,563	£ 1,110,563	£ 1,306,544	£ 1,413,829
>95% polyester	£ 1,005,084	£ 1,005,084	£ 1,182,452	£ 1,279,547
polycotton	£ 378,221	£ 378,221	£ 444,965	£ 481,503
>95% acrylic	£ 143,316	£ 143,316	£ 168,608	£ 182,453
>80% cotton	£ 293,109	£ 293,109	£ 344,835	£ 373,150
>80% wool	£ 624,944	£ 624,944	£ 735,228	£ 795,600
Other	£ 230,932	£ 230,932	£ 271,685	£ 293,994
Total	**£ 3,786,169**	**£ 3,786,169**	**£ 4,454,316**	**£ 4,820,075**
Cost of goods sold				
Transport costs	£-3,442,500	£-3,442,500	£-3,442,500	£-3,442,500
Purchase of input material	£-2,500,000	£-2,500,000	£-3,000,000	£-3,750,000
Labour	£-1,406,340	£-1,406,340	£-1,406,340	£-1,406,340

EOL treatment	£-1,782,969	£-1,782,969	£-1,603,493	£-1,505,244
	£-4,131,809	**£-9,131,809**	**£-9,452,333**	**£-10,104,084**
Gross profit	**£-345,640**	**£-5,345,640**	**£-4,998,017**	**£-5,284,009**
Operating expenditure				
Overheads	£-634,725	£-634,725	£-634,725	£-634,725
Facility Costs	£-1,111,486	£-1,111,486	£-1,111,486	£-1,111,486
Repairs & Maintenance	£-504,000	£-504,000	£-504,000	£-504,000
	£-2,250,211	**£-2,250,211**	**£-2,250,211**	**£-2,250,211**
EBITDA	**£-2,595,852**	**£-7,595,852**	**£-7,248,228**	**£-7,534,221**

5.0 Circle-8 Textile Ecosystems

Circle-8 are an ACT UK consortia member that have committed to developing a comprehensive, market-ready and investible business case for the first UK 25,000 T/a ATSP. Key points below.

Plant design

- Circle-8 are planning to build a fully automated facility designed to sort and pre-process NRT, to deliver feedstock for chemical and mechanical recycling.
- TOMRA sorting technology will be combined with other sorting and processing equipment as part of the full design layout for cutting, shredding and removal of seams, zippers and metals to enable 'pure textile output fractions' as feedstock for recyclers.

Supply, offtake and logistics

- A blended post-use supply approach, combining post-use commercial, business to business textiles and PCT, with RT and NRT pre-sort partnerships.
- Committed ATSP outputs of sorted and pre-processed cotton, polyester and polycotton for chemical and mechanical recycling in the UK and global.
- Greater emphasis in initial years on existing UK and EU open loop recyclers, as chemical recycling scales.
- Partnerships to support bringing down logistics costs and bringing up environmental efficiencies to NRT materials flows.

Brand engagement

- Major UK brand commitments to engage in supply via advanced collection systems and in-direct sales demand for recycled content.
- Exploring collaborative commitments to contributing to NRT handling through a gate fee to provide ecosystem investment prior to EPR, gaining competitive advantage for pro-active brand and retailers in the process. To be phased out as EPR comes into effect.
- A key principle is that ATSPs will not take on the cost burden of low value, unrecyclable fractions.
- As mechanical and chemical recycling industries mature, gate fee adjusted to reflect supply of increased value of recyclable materials in relation to the costs of non-recyclable materials.

Financial

- £15-22m CAPEX.
- Automation at scale, high volumes and industry-wide continued cooperation to achieve ecosystem economics is needed to incentivise involvement of all parties.



Figure 12: TOMRA automated sorting technology for textiles

6.0 Opportunities and conclusions

ATSPs must overcome two primary challenges to ensure long term viability and support textiles circularity. First, they require capital investment to purchase necessary technology and machinery. Second, they need financial support to break even and run operations whilst ATSPs are in their infancy, pursuing open-loop whilst the closed-loop recycling market develops and matures.

Strengthening the business case

This report has shown that the ATSP in the cost model does not create positive earnings at full capacity. Whilst the scenarios modelled are limited to available data and well-considered assumptions, pain points in the model highlight opportunities to leverage and build a strong, positive business case, outlined in Table 9.

Table 9: Opportunities for improving the ATSP business case.

Opportunity	Description
Logistics and transport costs	Streamline logistics, improve data or utilise logistics from existing business operations – see ATSP configurations 3.1.
Location	Strategic positioning to key NRT suppliers, feedstock buyers and transport infrastructure e.g., ports.
Staffing	Form synergies with existing solution providers, upskill or further automate workforce.
Facility equipment	Optimise set up for sorting accuracy and capability. Select technologies that overcome low acceptance rates relating to multi-layers and disruptors.
Acceptance rate and input material	Request offtake-agreements with suppliers specifying desired quality standards.
Output material	Respond to recycling market needs, this includes open and closed-loop recycling.
Rejected volumes	Find alternative onward markets to EOL for volumes rejected by ATSPs to both avoid gate fees and landfill tax as well as gain revenue from other markets.
Unlocking ATSP configurations	Integrate an ATSP into existing facilities or develop ATSP partnerships to utilise NRT supply, lower transport costs, staffing and CAPEX.
Service business model	Consider a service business model - see 3.1. Costs relate to operations only e.g., labour, maintenance, repair, facility. ATSPs need to charge customers an estimated minimum of £163 per tonne to break even.

Key enablers

Without external financial investment, ATSPs cannot cover costs even when operating at full capacity under the current cost model. Given the significant opportunity to support a circular textiles ecosystem and bridge the gap between NRT supply and textile recycling, support from key enablers is essential. Enablers can signal the demand for textiles recycling and therefore sorting for recycling; vitally de-risking investments and accelerating implementation.

Government support

Textile policy will support the successful implementation of ATSPs - see 'Transitioning to a UK circular textiles ecosystem' report[1] for policy recommendations. Appropriate EPR fees in particular could relieve ATSP operational costs. Government grants, public funding and regional subsidies can help ease the CAPEX bill. CAPEX grants can also incentivise private investors to engage in fundraising rounds.

Green investment

Investment in ATSP technology and infrastructure by impact investors, private equity, and venture capital can substantially reduce upfront costs and encourage further investment in the landscape. Investment is expanded upon in 'Transitioning to a UK Circular Textiles Ecosystem'[1].

Commercial partnerships

Brands and retailers have ambitious environmental targets to achieve, such as Net Zero[12] and UK Textiles Pact[11]. Recycled content is a key route to reach these targets. However, the level of recycled content required to meet these targets[1] is highly ambitious and unrealistic without strategic, bold and impact-led commercial partnerships. Forming direct supply partnerships that meet desired quality standards and contributing to the cost of NRT handling can reduce ATSP costs for pre-processing materials. Particularly at this critical stage, where ATSPs and closed-loop recycling markets are developing, and in the absence of textiles policy like EPR. Partnerships will help to ensure a steady supply of recycled content, mitigate risk in recycled content strategies and minimise textile volumes reaching landfill and incineration. Brands and retailers can also support ATSPs by creating effective take-back schemes with charities and collectors to guarantee NRT streams to ATSPs, ensure demand for recycled content through offtake agreements and letters of intent with recycling suppliers, and by designing products for recyclability, thereby supporting the textiles ecosystem across the value chain.

A supportive wider textiles ecosystem

The success of an ATSP network requires every part of the value chain to work collaboratively with the shared aim of creating a circular textiles ecosystem that is effective and future proof. Textile collections must be fit for purpose to increase NRT collected, divert EOL and foster positive citizen disposal behaviours, ensuring ATSPs secure reliable and quality input material. Please read the 'In-focus: Post-consumer textiles collections' ACT UK report[22], which spotlights collections as an emerging area for innovations in textiles circularity. Our comprehensive report, 'Transitioning to a UK Circular Textiles Ecosystem'[1] also builds on research from this paper, providing a macro perspective on how UK-wide ATSP networks can bridge critical gaps in achieving a textiles circular economy. It further emphasises the scale of opportunity ATSPs hold and the economic, environmental, policy and social enablers necessary for systemic change.

7.0 Glossary

ATSP (Advanced Textile Sorting and Preprocessing Facility): An optimised industrial sorting and pre-sorting process where individual specifications for categorisation can be customised with advanced technology, incl. state of the art optical scanning, robotics, AI, pre-processing (buttons, zips, trim removal) and size reduction equipment.[23]

Brands (surplus and unsold stock, some level of returns): Seller of textile stock that is yet to be used by a consumer or transferred to a retailer. As pre-consumer generators, brands generate textile resources in the form of surplus stock, unsold stock, returns, and damaged or faulty products.

Brands (take-back schemes): Producers of textile stock that has been purchased, used and discarded for reuse or disposal. As a post-consumer collector, they apply take-back schemes to collect the textiles after they have been discarded.

Charity retailer (in-store collections): Charities that collect (used) textiles via donations from citizens and corporations and/or through retailer take-back schemes in-store and/or online to fund charitable objectives.[24]

Charity retailers (charity shops): Charities that sell collected (used) textiles via donations from citizens and corporations directly and/or through retailers take back schemes in-store and/or online to fund charitable objectives, generally through the (re-)sale of collected items.[24]

Collectors: Commercial organisations that collect pre-consumer, post-industrial and post-consumer textiles. This general term includes stakeholders such as brands (take-back schemes), local authorities, commercial collectors, and charity retailers. Often, these organisations are the commercial collection partners for local authorities and retailer take-back schemes. They may also be charity-owned not-for-profit businesses for the sole purpose of generating funds for the charity through the sale of part of the collected items. In some cases, such as 'Cash for Clothes', the consumer gets a cash benefit if they give their clothes directly to the commercial collector. Some commercial sorters and collectors function as a single business operation.[24] Based on WRAP survey results, the assumption in this report is that 60%[2] of PCT collecting is managed by Local Authorities, 22% by charity collections and 30% by commercial collectors.

Closed-loop chemical recycling: Refers to the process where materials are taken back to their polymer (includes pulping, solvent-based and hydrothermal processes) or monomer level (including Methanolysis, Glycolysis, Hydrolysis and Enzymatic processes). Considering that it is framed in a closed loop, it creates outputs for use within the same industry.[24]

Closed-loop mechanical recycling: Refers to the process where materials are mechanically processed to recover and be used in new textiles (includes processes such as cutting, shredding and flocking). Considering that it is framed in a closed loop, it creates outputs for use within the same industry.[24]

Commercial laundries: Refers to businesses that provide large-scale washing and cleaning services for textiles.

End market: Refers to the final destination where discarded textiles are utilised or sold. In this stage end markets include charity retailers (charity shops), Peer-to-peer reuse organisations, international reuse markets, open-loop recycling, closed-loop mechanical/chemical recycling, and landfill or incineration.

End of life: Refers to the final stage in the lifecycle of a product, when the product no longer satisfies the last user.[25] At this point the product becomes part of the waste stream and will be handled by formal and/or informal waste management operators.

Garment manufacturing sector in EU: Manufacturers established in the EU, who generate fibres, yarns, fabrics and create products from the textile manufacturing process. These products can either be finished materials/products that are suitable for use/sale, or unusable materials and by-products.[24]

Garment manufacturing sector in the UK: Manufacturers established in the UK, which generate fibres, yarns, fabrics and products from the textile manufacturing process. These can either be finished materials/products that are suitable for use/sale, or unusable materials and by-products.[24]

Incineration: Waste treatment process that involves the combustion of substances contained in waste materials.

Individual consumers: Individuals who purchase, use, and then discard, resell or donate textiles.

Landfill: Refers to facilities that manage the disposal of solid waste.

Local Authorities (LA) (kerbside collections, textile bins, local community schemes): May provide textile collections as a service. Collections can include separate textiles at the kerbside, textiles bring banks at LA-managed bring sites and Household Waste Recycling Centers (HWRCs), as well as bookable home collection services. LAs commonly outsource their collection services to commercial collectors/sorters.[24]

Near-infrared (NIR) spectroscopy: Is a method used to identify fibre composition of textile products by measuring how much light a sample absorbs in the near-infrared region of the electromagnetic spectrum.

Open-loop recycling (wipers, insulation, shredding, composites or bricks): Refers to the recycling of materials where inputs from one industry are recycled into outputs for another industry[24], e.g. Shred Station in the UK.

Peer-to-peer (P2P) reuse organisation: (Online) platforms that enable consumers to transfer/sell textile products directly to other consumers, for example, eCommerce platforms, that enable Peer-to-Peer (P2P) buying and selling[24], estimated to be worth 46 million GBP annually.[26]

Post-consumer textiles (PCT or Post-CTs): Refers to the textiles that have been purchased, used and then discarded for reuse or disposal. The producers of these textiles include individual consumers, commercial laundries, brands (deadstock and take-back schemes) as well as uniformed institutions.

Post-industrial textiles (PIT): Refers to production waste, which is generated at the manufacturing level, and encompasses sliver cuts, garment cutting excess, trimmings, print trials and errors in dye lots, yarn waste, end rolls and rejected materials among others.[27]

Pre-consumer textiles (Pre-CTs): Refers to textile products that were manufactured, produced and intended for sale to the consumer or commercial/industrial end users, but remain unsold for reasons, such as overproduction (unsold stock/deadstock), cancelled orders or damaged/faulty stock. This also includes returned stock; products that have been returned by the consumer, as these have not been used.[29] It is estimated that pre-consumer waste at the retail phase in Europe accounted for 200,000T[1414] in 2020 and expects this to increase to 300,000 T by 2030. The survey run by WRAP for this project received only a small number of responses, with varying degrees of detail, reiterating the reluctance within the textile industry to be transparent on retail phase textile waste. Therefore, quantitative data cannot be aggregated to provide a representative figure for the UK.

Pre-consumer generator: Refers to the producer of textile stock that has been manufactured for and transferred to a retailer but is yet to be used by a consumer. This can include brands for unsold products, returns, and damaged or faulty goods.

Professional sorters: Commercial organisations that purchase or receive donations of textiles from other stakeholders, such as local authorities or charities, to sort into quality grades for reuse/recycling. Some commercial sorters and collectors function as a single business operation, some are charity-owned.[24]

Reuse export brokers: Organisations that may export used textiles for their original, intended purpose, direct reuse, or facilitate the buying and selling of used textiles from within/outside the UK. This may not include the physical handling of used textiles, for example, arranging a shipment or transaction.[2] The total value of used textile exports registered by the United Kingdom in 2023 amounts to $486,859,583[28], with the top importing countries by volume being: the United Arab Emirates (value: $62,229,885, volume: 73 kt), Poland (value: $81,642,539, volume 62 kt) and Ghana (value: $74,911,297; volume: 55 kt).

Uniformed institutions (workwear): Establishments where employees/members wear uniforms, contribute to the textile waste stream by disposing of uniforms and workwear that are worn out, outdated, or no longer fit for use.

Waste management companies: Organisations that provide services for the collection, transportation, processing, and disposal of (mixed) waste materials, including textiles

8.0 Appendix

8.1 Appendix 1: NRT profile methodology

Surveys and interview scope

A survey[29] was conducted as part of the ACT UK project to inform the profile of NRT flows in the UK. This included both online surveys (7) and extended telephone interviews (5) facilitated by the Textile Recycling Association (TRA).

The survey explored the landscape of textile collectors and sorters within the UK to understand capacity, costs, impacts and capabilities, particularly in terms of NRT flows. The survey specified interest in NRTs and any aspects of textile collection and sorting, including RTs, which could affect NRT logistics, flows and costs.

Respondents
12 respondents were part of the sample, including commercial collectors (used textiles, bank and charity retail, working with retailers), commercial sorters (including grading), pre-processors, used textile sellers (e.g. charity retailers), remanufacturers (repair, reuse and recycling), transportation and destruction.

Textiles scope
Clothing (excluding workwear), workwear, footwear, home textiles, industrial textiles (hospitality, hotels, commercial etc.), accessories, other.

Question Design
The survey included, but was not limited to, questions surrounding:

- The selling and buying of NRTs, both sorted and unsorted.
- Collection methods and grading for sorted NRTs
- Source and onward destination of textiles handled
- Whether businesses sort in the UK
- The approximate annual tonnage of used textiles handled by organisations
- The types of textile collected by an organisation
- Costs and observations associated with logistics, operations, transportation and overheads.

Limitations and assumptions
There are uncertainties around the percentage of NRT vs RT in many of these streams – in some cases an approximation has been made based on qualitative information and/or a proxy used.

This report is based on several informed assumptions due to a lack of robust data in these areas:

- NRT fibre composition is the same composition as textiles sold in the UK in 2019

- Assumes 15% of cotton and polyester blends, and 50% of wool is within blends, that are not recyclable by fibre-to-fibre recycling
- All post-consumer textiles sold to the UK public through charity retailers, peer-to-peer reuse organisations and used textile retailers are RTs
- Diverted from EOL and those collected by commercial collectors/sorters/charities are 34% NRTs
- 75% of textiles sent for EOL via residual waste are NRTs
- HWRC, kerbside and bring banks sent for EOL by LAs are 60% NRT
- All rags and those sent for recycling are NRTs.

Sensitivity analysis

There are several factors contributing to the uncertainty in NRT and RT split within residual waste:

- Very limited data available
- Subjective nature of determining NRT or RT after it has been contaminated in residual waste
- Uncertainty in condition of textiles before entering residual waste stream.

In addition to this, a large proportion of post-consumer textiles enters residual waste, and therefore there is large potential for this to become feedstock for an ATSP and fibre-to-fibre recycling system.

It is therefore important to check the sensitivity of the NRT flows to changes in the % NRT in residual waste assumption.

Table 10: Sensitivity analysis on % NRT in residual waste, resulting NRT tonnes generated and approximate total NRT with F2F recycling potential.

% of NRT in total residual waste	Total NRTs generated by citizens 1000 tonnes / yr
50	585
75	743
85	851
90	889
95	927

8.2 Appendix 2: Current and future recycling capacity overview

Table 11: Summary of closed-loop recycling capacities as indicated in WRAP's Textile Recycling Database (status: 2023).

	Mechanical	Chemical
	Based on information from 64 recyclers	Based on information from 38 recyclers including 3 recyclers both mechanical and chemical
Current Capacity	- 3 at pilot scale; - 3 at scale with post-industrial waste and piloting post-consumer waste; - 27 at scale (no capacity given); - For the rest, the capacity at scale ranges from 50 T/a to 780 000 T/a or 136,000 T/a of waste processed	- 4 recyclers did not provide scale; - 4 demo plants; - 11 pilot/research scale; - 12 at scale (no capacity given); - For the rest, the capacity ranges from 264 T/a to 60,000 T/a
Future Capacity (2025 - 2030)	- 2 recyclers intend to maintain their capacity - 15 recyclers have provided future capacity which ranges from 800 T/a to 36,000 T/a	- 11 recyclers have provided future capacity which ranges from 5,000 T/a to 500,000 T/a
Input (post-industrial pre-consumer post-consumer)	- 23 recyclers of post-industrial - 18 recyclers of post-industrial and pre-consumer - 11 recyclers of all inputs (including mattresses and footwear) - 10 recyclers of post-consumer - 1 recycler of pre-consumer	- 10 recyclers of post-industrial - 2 recyclers of post-industrial and pre-consumer - 8 recyclers of all input types - 8 recyclers of post-consumer - 9 recyclers of post-consumer and post-industrial - 1 did not specify
Fabric structure	Only 8 recyclers indicated the types of fabric structures accepted - 4 recycling all structures (including shoes) - 1 recycling only knits - 3 recycling knits and wovens (including the recycler with the largest current capacity volume)	Only 8 recyclers indicated the types of fabric structures accepted - 5 recycling all structures - 1 recycling only their own products - 2 accepting knits and wovens
Materials recycled (input)	- 38 list cotton as the primary material - 12 list wool as the primary material - 21 recyclers are highly specialised and accept only one material type: down, leather, wool, acrylic, aramid, cashmere, cotton, polyamide, or polyester - 18 recyclers accept blends (acrylic polyester, acrylic polypropylene, cotton blends, polycotton blends, poly elastane blends and more) - 1 focused only on footwear	- 12 list cotton as the primary material - 11 list polyester as the primary material - 22 recyclers are highly specialised and accept only one material type: cotton, elastane, leather, nylon 6, nylon 6.6[25], polyester - 7 recyclers accept blends (mainly cotton blends and polycotton) - 5 did not indicate materials

8.3 Appendix 3: ATSP cost model assumptions

Table 12: ATSP cost model assumptions.

Item/Metric	Input	Source	Notes
TEXTILE HANDLING COSTS			
Collection costs	£214 per tonne	Textile Market Situation Report, p.25[26]	
Transport & Logistics	£138 per tonne	Aggregated Sorters Survey Data[29]	
Processing costs	£175 per tonne	Aggregated Sorters Survey Data[29]	
Total UK sorter volume	469,000 tonnes	Textile Market Situation Report, p.20[26]	
Estimated split of NRT in current collections	32%	Aggregated Sorters Survey Data[2929]	
Average NRT Value	£100 per tonne	Aggregated Sorters Survey Data[2929]	
EfW gate fee	£112 per tonne	WRAP Gate Fees Report	
Non-hazardous Landfill gate fee	£24 per tonne	WRAP's seventeenth Gate Fees Report	
Non-hazardous Landfill tax (£ per tonne)	£104 per tonne	Landfill Tax Rates Gov.UK	
MATERIAL INPUT PRICES			
Scenario 1 - input material is a revenue stream, all quality accepted	£ 100		
Scenario 2 - input material is a cost, all quality accepted	£ (100)		
Scenario 3 - input material is a medium cost, only single layered quality	£ (120)		
Scenario 4 - input material is a high cost, only single layered quality and disrupter criteria	£ (150)		
CAPEX			
Total Capital Expenditure	£ 7,000,000	Valvan	
CAPEX contingency	20%	Industry standard	
CAPEX upgrade	20%	Industry standard	
FACILITY COSTS			
Item/Metric	**Input**	**Source**	**Notes**
Ground rent	£8.30 per sq ft per year		Industrial facility average ground rent based on Circle-8 site cost analysis

Security deposit	£ 373,500	Legal Foundations	6 months of rent
Stamp duty	£ 48,500	Stamp Duty Land Tax	Set by the government
Maintenance & repair	5% of RAV (Replacement Asset Value)	Upkeep.com	
Business Rates	£3 per year		Based on Circle-8 site cost analysis for WP 6
Environmental monitoring	£8,000 per year	Costs for Municipal Waste Management in the EU	MRF UK (no textiles) for a capacity of 42,000 tonnes
Gas needs	30,000 kWh	Average business gas and electricity bills explained	Low end usage for a medium size business
Gas Price	£0.08 per kWh		Price for a medium size business
Gas standing charge	£0.48 per day		Price for a medium size business
Electricity needs	25,000 kWh		Low end usage for a medium size business
Electricity price	£0.254 per kWh		Price for a medium size business
Electricity standing charge	£0.88 per day		Price for a medium size business
Water usage	50 cubic meters per employee		
Water costs	£2 per cubic meter	Business Water Rates 2024/25	
Wastewater costs	£2 per cubic meter		
Insurance	£20,000 per year	Key Expenses in Recycling	From discussions with sorters
Miscellaneous other costs	£0.19 per sq ft per year		From discussions with sorters
OPERATING INPUTS			
Starting capacity	10,000 tonnes		First years capacity
Maximum capacity	25,000 tonnes	Circle-8	From interview with Circle-8
Depreciation term	15 years		Standard depreciation term
Ramp up period	3 years		Currently assumed ramp up time
Item/Metric	**Input**	**Source**	**Notes**
Start-up costs	£250,000 per year		Extra costs to incur during the ramp up process, such as training, travel, project management, from discussions with sorters
Year 2 start-up cost vs	50%		

Year 1			
Shifts	3 shifts per day		From discussions with sorters
Warehouse size	40,000 sq ft		Based on Circle-8 site cost analysis for WP 6
Total line size	50,000 sq ft		Valvan
Number of lines	7	Sorting for Circularity Europe (2022)	Based on 6 recycling grades: Chemical Recycling: Polyester, Chemical Recycling: Polycotton, Chemical Recycling: Cotton, Mechanical Recycling: Cotton, Mechanical Recycling Acrylic, Mechanical Recycling: Wool, Variable line
Total disruptor removal line	5		Based on 6 recycling grades: Chemical Recycling: Polyester, Chemical Recycling: Polycotton, Chemical Recycling: Cotton, Mechanical Recycling: Cotton, Mechanical Recycling Acrylic, Mechanical Recycling: Wool, Variable line
Shift hours	1920 hours per shift per year		Assuming 40 hours per week, 48 weeks per year
Operating days per year	260 days		Assuming operations 5 days a week all year round

Table 13: Staff required to operate the ATSP as a stand-alone facility.

Job Title	Team	Job Description	Number	Base salary
Managing director	Management	Directs and controls the company's operations and gives strategic guidance and direction to the board to ensure that the company achieves its mission and objectives	1	£ 63,000
Production manager	Operations	Ensures that the production starts and finishes within budget and on time and ensures all products meet the required quality standards	3	£ 42,000
Production worker	Operations	Responsible for the automated sorting process from bale opening to disrupter removal (full flow). This includes machine operations and quality assurance and quality control procedures	3	£ 29,000
Warehouse manager	Operations	Responsible for organising deliveries, storage and despatches of goods in the warehouse, oversees the warehouse team.	3	£ 45,000
Environmental and industrial risk advisor	Operations	Providing environment and industrial risk compliance support for all areas of the business	1	£ 36,000

Engineer	Operations	Provide planned and reactive maintenance for all areas of plant, making sure plant installations and modifications meet the latest standards and regulatory requirements	1	£ 37,000
Warehouse worker	Operations	Loading and unloading of freight, picking and packing products, parcel sorting, baling, general warehouse duties	3	£ 27,000
Bale press worker	Operations	Managing the bale press, general warehouse duties	1	£ 27,000
Sales agents	Sales	Responsible for selling products and meeting customer needs while generating feedstock volumes through existing and new channels	1	£ 34,000
Business development manager	Sales	Develops, manages and maintains a pipeline of new buy/sell business opportunities, works on innovative projects	1	£ 48,000
HR manager	Management	Manages talent, compensation and benefits, training and development, compliance, and worker safety	1	£ 37,000
Logistics manager	Operations	Oversees and plans the movement and storage of supplies within the facility	1	£ 48,000
Financial controller	Management	Maintaining internal controls over financial reporting, presenting financial reports to an executive team and overseeing the preparation of budgets	1	£ 55,000
Bookkeeper	Administration	Responsible for recording and maintaining business' financial transactions, such as purchases, expenses, sales revenue, invoices, and payments	1	£ 28,500

9.0 Endnotes

[1] WRAP. (2025) *Transitioning to a UK circular textiles ecosystem.*
[2] WRAP. (unpublished). *Current non-reuseable textiles flows in the UK*. Data has been collated from WRAPs Textile Market Situation Report (2024), an ACT UK workshops (2024), Sorters Survey (2024) (conducted by the Textiles Recycling Association on behalf of WRAP) and follow up interviews (2024).
[3] WRAP. (unpublished) *'Projection of Flows of Non-Reusable Textiles Scenarios UK'*.
[4] Numbers used in the NRT and ATSP Cost Model are obtained from WRAPs Textile Market Situation Report (2024) and Sorters Survey (2024) (conducted by the Textiles Recycling Association on behalf of WRAP) and follow up interviews (2024). Collection costs are reported in the Textiles Market Situation Report (p.25), total PCT sorted volume in the UK from the Textiles Market Situation Report (p.20). The following assumptions are based on the survey data: Transport & Logistic costs, processing costs, estimated share of NRT within the total PCT collected, and average NRT value. The full list of assumptions applied in ATSP and NRT cost modelling can be found in Appendix 3.
[5] Fashion for Good. (2023) *Sorting for circularity: India: Annex 2 Assumptions for the Business Case Analysis*.
[6] Fibersort. (n.d.) *Fibersort technology*.
[7] Solstice. (n.d.) *Meet the SOLSTICE partners*.
[8] Matoha. (n.d.) *Fabrics Identification*
[9] MTC. (2025). *Condition of supply pre-processing technology assessment*
[10] WRAP. (2024) *Textiles Sorting And Recycling Database*.
[11] WRAP (n.d.) *UK Textiles Pact*.
[12] Gov.uk. (n.d.) *Net Zero*.
[13] European Commission (2024). *Ecodesign for Sustainable Products Regulation*.
[14] McKinsey & Company. (2024) *Scaling textile recycling in Europe-turning waste into value.*
[15] This includes 2 UK and EU businesses, representing the majority of predicted 2030 volumes, with the biggest predicted 2030 capacity by ISKO. This excludes Remanufacturers who are not considered potential clients of an ATSP, but rather competitors.
[16] GeoMapTric. (2024) *Special Economic Zones - England.*
[17] EigenDraads (2022). Feasibility of Rotterdam region as feedstock supplier for fibre-to-fibre recycling. Available upon request from Fashion for Good.
[18] Sales prices are similar to the feedstock output prices quoted in the Fashion for Good. (2022) Sorting For Circularity Europe: An Evaluation And Commercial Assessment Of Textile Waste Across Europe, which are based on data obtained from textile sorters. Note, however that the ATSP cost model developed for this project offers the ability to perform separate pricing sensitivity analysis.
[19] In the ATSP cost model, costs of colour sorting are excluded, and met by mechanical recyclers.
[20] EigenDraads (2022). *Business case for automated sorting and hardware removal in the Rotterdam region.*
[21] Evidence based from Fashion for Good. (2022) Sorting For Circularity Europe: An Evaluation And Commercial Assessment Of Textile Waste Across Europe. From stakeholder expert interviews for this ACT UK report, it is assumed that plastic and metal disruptors when removed would lower garment weights by 20%, while textile and other disruptors would lower the garment weight by 50%. It was assumed that textiles and other disruptors would likely not be picked out by the NIR sorting technology as the disruptor would cover the main fabric.
[22] WRAP. (2025) *In-focus: Post-consumer textiles collections.*
[23] UKFT. (2023) *Developing a Pioneering Pre-Processing Demonstrator for Waste Textiles*.
[24] WRAP. (2024) *Textiles Reuse and Recycling Definitions.*
[25] Alamerew, Y.A., Brissaud, D. (2019) *Circular economy assessment tool for end-of-life product recovery strategies.* Volume 9, Journal of Remanufacture, 169–185.
[26] WRAP. (2024) *Textiles Market Situation Report 2024*.
[27] WRAP. (2024) *Analysis of Waste Arising in the Textiles Life Cycle: Textile Waste Hotspots*.
[28] UN Comtrade Database. (n.d) *Trade Data*.
[29] WRAP. (2024) Sorters Survey (conducted by Textiles Recycling Association on behalf of WRAP).